Bill of Materials (BOM)

Hardware Architecture & Component Sourcing

shipping is free from robu.in for orders above 1k

For contigency, i took 25% of original total

Line Total	Shipping Cost	Contingency	Total Budget	Total Quantity
₹6,597.10	₹200.00	₹1,649.28	₹8,446.38	32

Item	Exact Model	Qty	Unit Price	Line Total	Supplier URL	Key Specs	Selection Reason	Alternative
Microcontroller	ESP32-WROOM-32 DevKit V1 (30-pin)	2	₹449.00	₹898.00	https://robu.in/product/esp	Dual-Core 240MHz, 520KB SRAM, 4MB Flash, 3.3V logic, Hardware PWM timers	Dual core handles simultaneous line-tracking PID and maze shortest-path memory calculation; 1 spare for pit repairs.	Raspberry Pi Pico W (Slower dual M0+ core, less RAM for node trees)
IR Array	5-Channel TCRT5000 IR Line Tracking Sensor Module	2	₹209.00	₹418.00	https://robu.in/product/tcrt	5-ch TCRT5000 array, analog & digital outputs, LM393 comparators, trimmers	Analog outputs allow software threshold calibration against glossy vinyl surface reflections.	3-Channel IR Array (Fewer sensors, unreliable 90° turn sensing)
Motors	GA12-N20 6V 500RPM Micro DC Reducer Motor Encoder with Wire	2	₹619.00	₹1,238.00	https://robu.in/product/ga1	6V DC, 500 RPM rated speed, all-metal reduction gearbox, integrated dual-channel incremental quadrature Hall encoder, 3mm D-shaft	500 RPM provides high linear speed to minimize actual run time, while quadrature encoder feedback guarantees precise 90° pivot turns without wheel slip	N20 6V 100RPM Encoder Motor (High torque but too slow to maximize the time score during the actual run)
Motor driver	TB6612FNG Dual H-Bridge Motor Driver Module	2	₹149.00	₹298.00	https://robu.in/product/tb6	VM: 4.5V–15V, 1.2A continuous / 3.2A peak, MOSFET H-Bridge, 3.3V logic	MOSFET design has negligible voltage drop (~0.2V) and runs cool compared to bipolar drivers.	L298N Module (Drops 2V as waste heat, excessively bulky)
Battery	Orange 7.4V (2S) 1500mAh 25C LiPo Battery Pack	2	₹1,239.00	₹2,478.00	https://robu.in/product/orai	7.4V nominal (8.4V max), 1500mAh, 25C discharge (37.5A burst), XT60	Provides stiff voltage without sagging under sudden motor stall/acceleration; 2 packs allow back-to-back testing.	9V Alkaline Battery (High internal resistance, collapses under motor draw)
Battery Charger	IMAX B3 AC Pro Compact Balance Charger for 2S-3S LiPo	1	₹349.00	₹349.00	https://robu.in/product/b3a	AC 100–240V input, 3 x 800mA dedicated balance charging per cell	Essential balancer to prevent cell overcharge and protect battery cycle life.	Standard DC 9V Adapter (Dangerous for LiPo, causes puffing/fire)
Voltage Regulator	SmartElex Mini MP1584 DC-DC 3A Adjustable Buck Module	2	₹105.00	₹210.00	https://robu.in/product/mp1	Input: 4.5V–28V, Output: Regulated 5.0V, 3A peak, 96% efficiency	Highly efficient step-down from 7.4V LiPo to clean 5V for ESP32 and sensor logic.	LM2596 Module (Much larger footprint and unnecessary chassis weight)
Chassis	Line Follower Chassis Compatible with N20 Motor with Clamp (3D Printed, ABS/PETG)	1	₹169.00	₹169.00	https://www.indianhobbyco	Dimensions: 130 × 129.4 × 3 mm, material: ABS/PETG, includes 2x custom N20 motor mounting clamps, front sensor extension plate	Ultra-lightweight and rigid structure dedicated for N20 motors; fits within the 220 X 220 X 220 mm size limit and includes direct clamp hardware.	2WD Acrylic Round Chassis Kit (Heavier, generic circular profile without dedicated N20 clamps)
Drive Wheels	5908 Series Silicone Tire Wheel for N20 Motors (Pair)	2	₹169.00	₹338.00	https://probots.co.in/silicon	59mm outer diameter, 8mm width, high-grip molded silicone tread, lightweight polycarbonate 5-spoke rim, direct press-fit for 3mm D-shaft N20 motors.	High-grade silicone tread maximizes ground friction on vinyl tape, preventing lateral drift/skidding during high-speed sharp 90° maze turns and rapid acceleration.	3Pi miniQ 42mm Car Wheel Tyre (Lower top speed, but slightly more compact).
Support Wheels	MINI 3PI Car N20 Caster Robot Ball Wheel	1	₹39.00	₹39.00	https://robu.in/product/mini	12mm stainless steel 304 ball, 22 × 15 mm enclosure, 10g total weight, 360° omnidirectional rolling contact.	Provides smooth omnidirectional 3-point support with zero floor drag, prevents scratching/marking on the arena floor (strictly avoids rule penalties/disqualification).	Small Steel Ball Caster Wheel (35mm base plate, slightly bulkier).
Main Power Switch	KCD3-2-101C SPST Rocker Switch (6A 250V AC)	1	₹13.00	₹13.00	https://robu.in/product/kcd	Single Pole Single Throw (SPST) ON-OFF rocker switch, rated up to 6A/250VAC, snap-in panel mounting.	Fulfills Techfest single main power switch rule; handles high discharge current (25C LiPo burst) without contact arc or voltage drop.	Heavy Duty Toggle Switch (Bulkier).
Main Power Wiring	22AWG High Quality Ultra Flexible Silicone Wire (Black)	1	₹18.00	₹18.00	https://robu.in/product/22g	Multi-strand tinned copper conductor, high-temp flexible silicone rubber insulation (-60°C to +200°C), 22AWG thickness.	Flexible silicone wire doesn't snap under chassis vibration and soldered connections won't melt during high current draw from LiPo battery.	Standard 24AWG PVC Hookup Wire (Brittle, melts easily during soldering).
Red End LED	Cree 5mm Ultra-Bright Red DIP LED (C503B-RAN)	1	₹28.00	₹28.00	https://robu.in/product/c-5	5mm clear round lens, 624nm wavelength, 32,900 mcd (ultra-bright output), 2.1V forward voltage.	Highly visible ultra-bright red output ensures judge confirmation on the 400mm x 400mm end zone box to earn the +5 points bonus.	Standard 5mm Diffused Red LED (Duller brightness).
LED Resistor	220 Ohm Carbon Film Resistor (Multicomp Pro 220Ω)	10	₹0.51	₹5.10	https://robu.in/product/mcp	220Ω resistance, ±5% tolerance, 125mW power rating, axial leaded.	Safely limits current from a 5V/3.3V microcontroller digital output pin to protect both the LED and MCU.	330 Ohm Carbon Film Resistor (Slightly dimmer output).
Jumper Wires (F-F)	10CM Female to Female Breadboard Jumper DuPont Cable (40 Pcs)	1	₹49.00	₹49.00	https://robu.in/product/10c	10cm length, 2.54mm pitch 1P-1P female header at both ends, 40-pin detachable ribbon cable.	Short 10cm length keeps chassis wiring neat without excess wire loops; ideal for connecting ESP32 to TB6612FNG & Buck module.	20cm Female to Female Jumper Wires (Longer, creates wire mess).
Jumper Wires (M-F)	Male To Female Jumper Wires 40 Pcs 10cm	1	₹49.00	₹49.00	https://robu.in/product/mal	10cm length, 2.54mm pitch 1P male pin to 1P female header, 40-pin detachable ribbon cable.	Needed for connecting male headers (like sensor arrays or MCU breadboard pins) directly to female module pins.	Male to Male Jumper Wires (Only used for breadboards).
Total		**32**		**₹6,597.10**				